NAME ☞

SCHOOL ☞

TEACHER ☞

Pre-Junior Certificate Examination, 2015

Mathematics

Paper 2

Ordinary Level

Time: 2 hours

300 marks

For examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3		13	
4		14	
5			
6			
7			
8			
9			
10		Total	

School stamp

Running total	

Grade

Instructions

There are 14 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times, you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if all necessary work is not clearly shown.

You may lose marks if the appropriate units of measurement are not included, where relevant.

You may lose marks if your answers are not given in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1 **(Suggested maximum time: 5 minutes)**

(a) Nick wishes to watch a film on television.
The film starts at 19:25 and ends at 21:10.

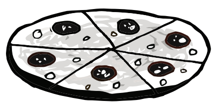
(i) How long is the film? Give your answer in hours and minutes.

(ii) Nick needs to complete his household chores before sitting down to watch the film.
The chores and the time needed to complete each one are shown in the table below.

Chore	Time (minutes)
Wash the dishes	25
Tidy his bedroom	20
Hang out the washing	15
Do the ironing	35

What is the latest time Nick can start his chores so that he can sit down 5 minutes before the film starts?

(b) Nick decides to order a takeaway pizza during the film.
The pizza has a diameter of 30 cm.
Find the area of the pizza, correct to the nearest cm^2.



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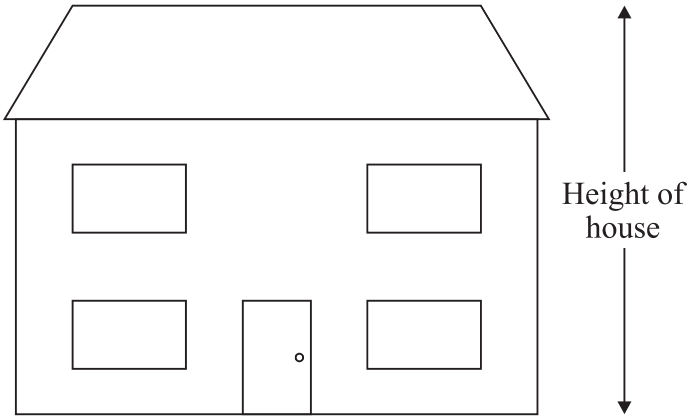
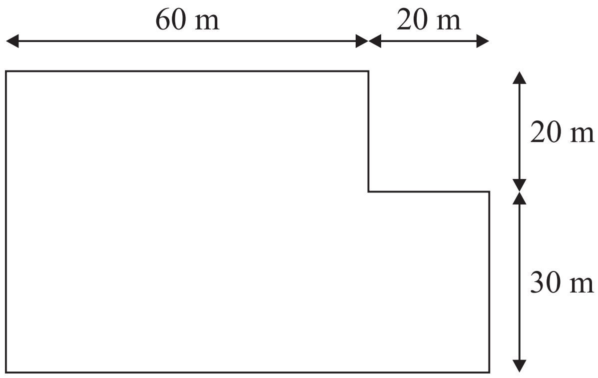
Question 2 **(Suggested maximum time: 10 minutes)**

(a) The diagram below shows a scale drawing of the front of a farmer's house.



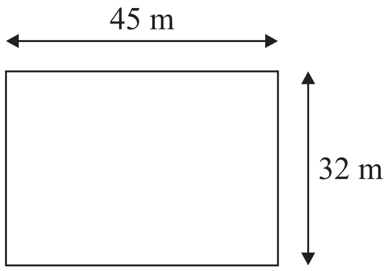
The actual height of the door is 2 metres.
Find the actual height of the house in metres.

(b) The farmer has a field with shape and measurements as shown in the diagram below.



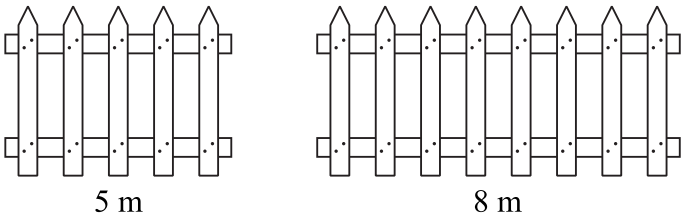
Find the area of the field, in m^2.

(c) The farmer keeps chickens in a rectangular yard with sides of length 32 m and 45 m as shown in the diagram below.



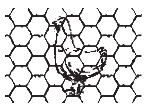
(i) Find the perimeter of the yard, in metres.

(ii) The farmer wishes to put a picket fence all the way around the yard before placing chicken wire on it. Sections of picket fence are sold in 5 m and 8 m lengths.



Find the least number of each section required to fence off the yard **without** the need to cut any of the sections.

(iii) The farmer can purchase chicken wire in rolls of 25 m length.
How many rolls does the farmer require to cover the fence?



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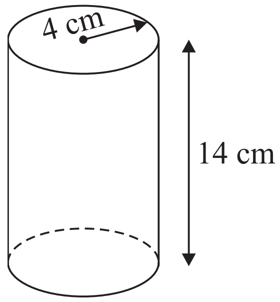
Question 3 **(Suggested maximum time: 10 minutes)**

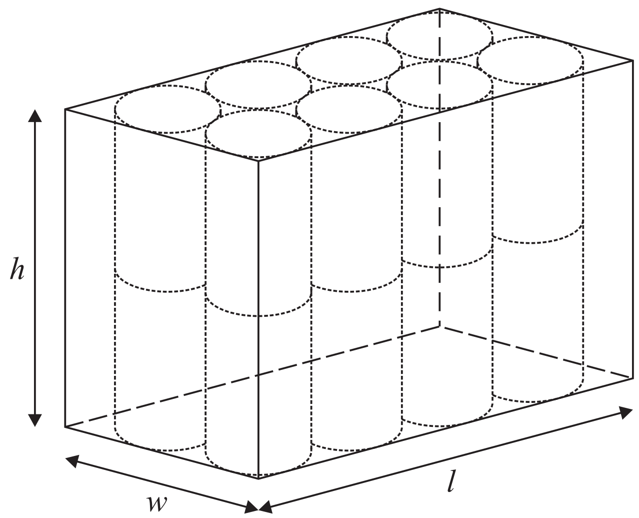
(a) Many foods are tinned to lengthen their lifespan. A food manufacturer produces soup in closed cylindrical tins.

A cylindrical tin has a radius of 4 cm and a height of 14 cm, as shown in the diagram.



Find the volume of a cylindrical tin. Take $\pi = 3{\cdot}142$.
Give your answer correct to the nearest cm^3.

(b) Sixteen tins of soup are packed tightly into a rectangular box, as shown.



(i) Find, in cm, the length, the width and the height of the rectangular box.

Length, l =

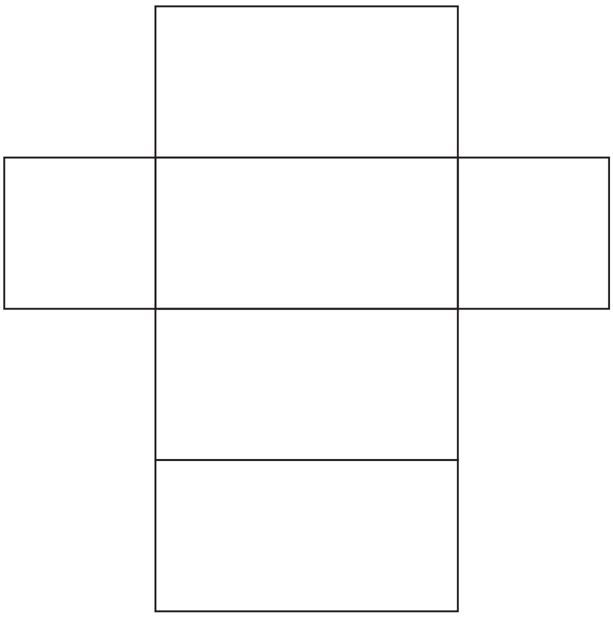
Width, w =

Height, h =

(ii) Find the volume of the rectangular box, in cm^3.

(iii) What percentage of the space in the rectangular box is **not** used?
Give your answer correct to one decimal place.

(iv) Thomas says that the net of this rectangular box is shown below.



Is he correct? Give a reason for your answer.

Answer:

Reason:

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Question 4 **(Suggested maximum time: 5 minutes)**

(a) Write down the name of the **metric** unit which is usually used to measure each of the following items. The first one has been completed for you.

Item	Metric Unit
Volume of milk in a carton	litres (l)
Weight of a person	
Height of a school building	
Weight of a chocolate bar	

(b) **(i)** Change 5 centimetres to millimetres.

(ii) Change 9400 millilitres to litres.

(c) John cycled 30 km in 1·5 hours. Jenny cycled 42 km in 2 hours.
Which cyclist had the greater average speed?

Question 5 **(Suggested maximum time: 5 minutes)**

Owen has a packet of flower seeds.
The probabilities that a seed will grow into a flower which is white, red, orange, yellow or pink are shown in the table below.

Colour	White	Red	Orange	Yellow	Pink
Probability	0·24	0·15	0·25	0·17	

Owen takes a seed from the packet at random and plants it in his garden.

(a) What is the probability that the seed chosen will grow into a flower which is pink?

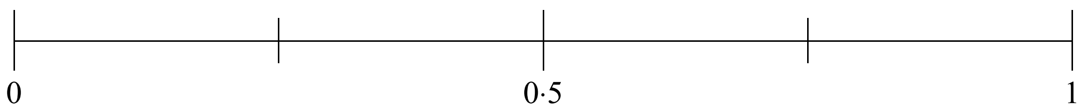
(b) What is the probability that the seed chosen will grow into a flower which is either red or pink?

(c) Another gardener tells Owen: "There will always be more orange flowers than red flowers produced from this brand of seeds".
Would you agree with this gardener? Give a reason for your answer.

Answer:

Reason:

(d) On the probability scale below, mark with the letter **X** the most appropriate position to show the probability that the seed chosen will grow into a flower which is **not** red.



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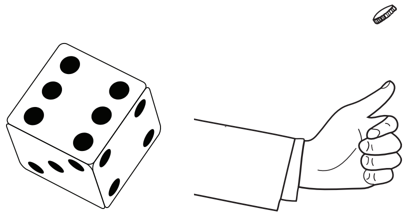
Question 6 **(Suggested maximum time: 5 minutes)**

A game is played using a die and a coin.
A player rolls the die and flips the coin at the same time.
Two possible outcomes are (**1**, **H**) and (**5**, **T**).



(a) Complete the table below to show all the possible outcomes.

	Head (H)	**Tail (T)**
1	**(1, H)**	
2		
3		
4		
5		**(5, T)**
6		

(b) How many possible outcomes are there?

(c) What is the probability that the outcome consists of an even number and a tail?

Answer =

(d) What is the probability that the outcome consists of an odd number, or a head, or both?

Answer =

(e) If Sam plays the game 90 times, how many times should he expect the outcome to contain a 6?

Question 7 **(Suggested maximum time: 10 minutes)**

The average retail prices for petrol and diesel in twelve European countries surveyed in June 2014 are shown in the table below. All prices are given in cent per litre.

Country	Petrol	Diesel	Country	Petrol	Diesel
France	151	138	**Latvia**	130	126
Germany	157	139	**Norway**	193	176
Great Britain	163	170	**Poland**	130	128
Greece	173	139	**Portugal**	156	132
Ireland	155	147	**Spain**	144	134
Italy	179	167	**Sweden**	165	159

Source: AA Ireland

(a) **(i)** Which country had the lowest average retail price per litre for diesel in June 2014?

(ii) Which country had the highest average retail price per litre for petrol in June 2014?

(b) One country had a different trend for petrol and diesel prices in June 2014 when compared to the other countries surveyed.
Name this country and explain the difference.

Country:

Explanation:

(c) Calculate the mean retail price for petrol in the countries surveyed in June 2014.

(d) Taxes on fuel are significantly higher in Norway.
If Norway were excluded from the survey, would you expect the mean retail price of petrol to be much higher or lower than your answer in part **(c)** above? Give a reason for your answer.

Answer:

Reason:

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Question 8 **(Suggested maximum time: 15 minutes)**

(a) Jane wishes to investigate the healthy eating habits of people in her local area. She plans to carry out a survey to find out the eating habits of local people. The answers to survey questions can be classified as:

- Discrete numerical data
- Continuous numerical data
- Nominal categorical data
- Ordinal categorical data

The table below shows some of the variables that Jane wishes to examine.
For each variable shown, identity the type of data it represents.

Variable	Type of Data
Favourite vegetable	
Age of person	
Number of healthy meals eaten per week	

(b) Jane chooses people at random for the survey and asks them to complete a questionnaire.
Write down two ways in which a survey using a questionnaire can be carried out.

First way:

Second way:

(c) Jane wants to find out which vegetable people like best.
One question on her survey is:

Question: What vegetables do you like?

(i) Explain why this is **not** a suitable question.

(ii) Rewrite the question in a suitable form.

(d) The stem-and-leaf diagram below shows the ages of the people who take part in the survey.

0					
1	5				
2	4	8	9		
3	0	0	1	2	8
4	2	3	3	3	
5	2	5			
6	1	5			
7	9				

Key: $2|4 = 24$

(i) How many people take part in the survey?

(ii) What is the **range** of the data?

(iii) What is the **mode** of the data?

(iv) Find the **median** age.

(v) The age of another person who is 51 years of age is added to the stem-and-leaf diagram. Find the **median** of this data.

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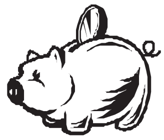
Question 9 **(Suggested maximum time: 5 minutes)**

Jean is saving money by placing her loose change in a piggy bank.
The coins she has saved are shown below.



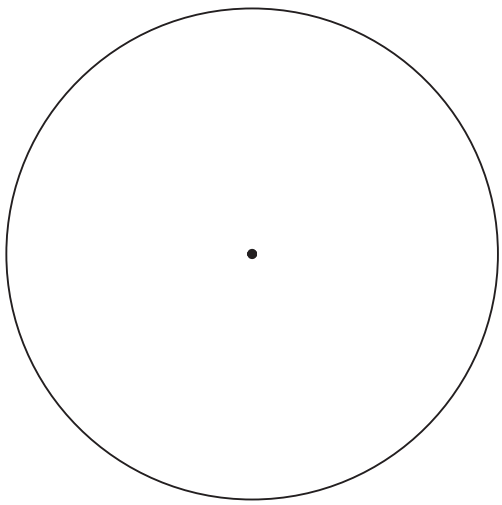
€2	50c	€1	€2	20c	50c
50c	€2	50c	20c	€2	€2
20c	50c	€2	20c	50c	20c
€2	20c	50c	50c	€2	€1

(a) Fill in the tally and frequency columns in the table below to show the money Jean has saved.

Value of Coin	Tally	Frequency
20c		
50c		
€1		
€2		

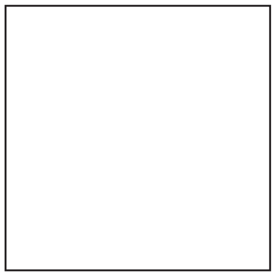
(b) Use your answers to part **(a)** to find the total amount of money Jean has saved.

(c) Complete the pie chart below to illustrate the data. Show all your calculations clearly.

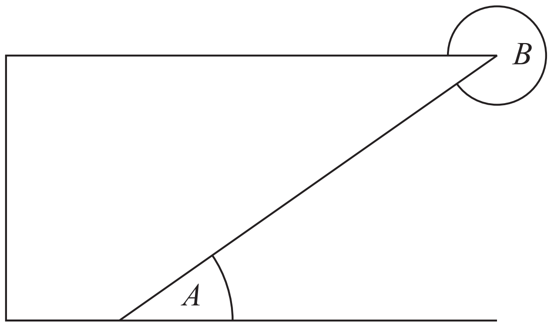


Question 10 **(Suggested maximum time: 10 minutes)**

(a) Draw **all** the axes of symmetry of the square shown in the diagram below.



(b) Two angles are marked on the diagram below.



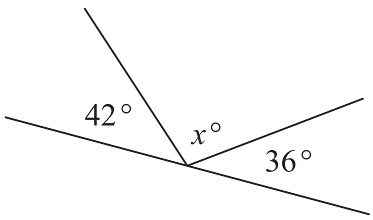
(i) Identify the **type of angle** which the angles A and B represent.

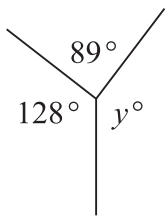
Angle	Type of Angle
A	
B	

(ii) Mark with the letter **R** a right angle in the diagram above.

(iii) Mark with arrows, (〉,〉), a pair of parallel lines in the diagram above.

(c) Find the measure of each of the missing angles in the diagrams below.
Show your calculations.





$x° =$

$y° =$

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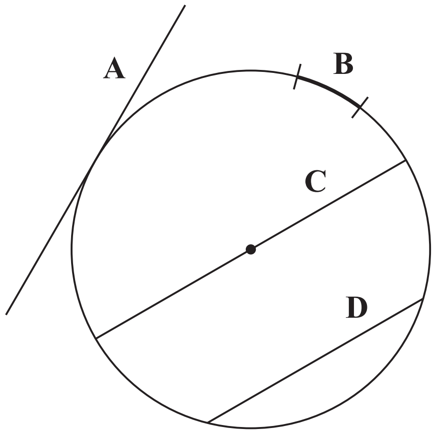
(d) The diagram below shows different elements of a circle.



Choose the correct term for each of the elements indicated in the diagram from the following table.

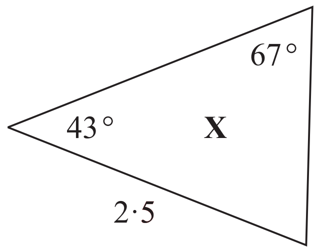
Diameter	Tangent	Arc	Chord	Radius	Circumference

Write the answers into the grid.

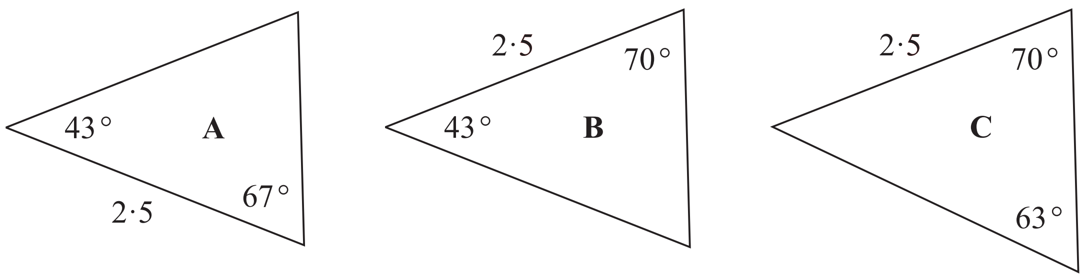
	Term
A	
B	
C	
D	

Question 11 **(Suggested maximum time: 5 minutes)**

(a) The triangle X is shown in the diagram below.



Which one of the triangles A, B or C below is congruent to triangle X?
Give a reason for your answer.



Answer:

Reason:

(b) Construct a triangle PQR with $|PQ| = 9{\cdot}5$ cm, $|QR| = 8$ cm and $|\angle PQR| = 40°$.
Label your diagram clearly.

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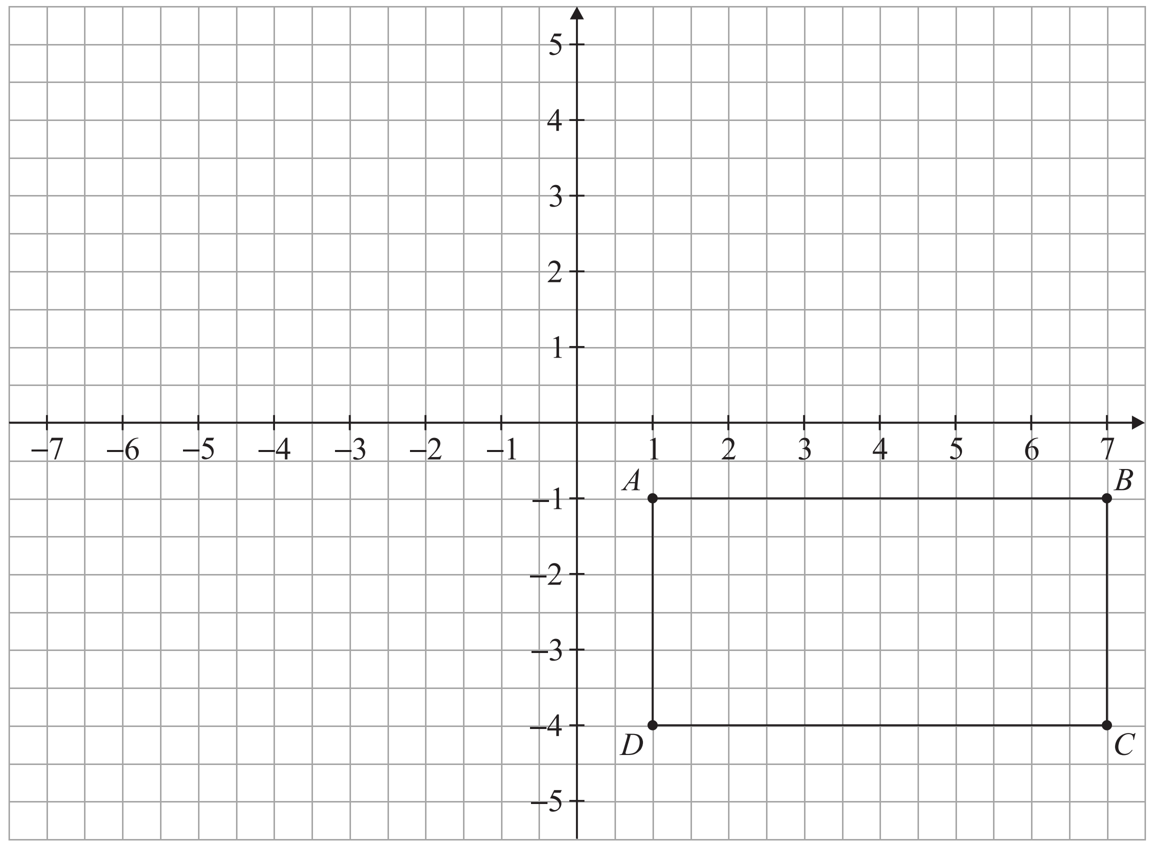
Question 12 **(Suggested maximum time: 10 minutes)**

The diagram below shows a rectangle on the co-ordinated plane.

(a) Draw the image of the rectangle under an axial symmetry in the y-axis.



(b) A, B, C and D are the vertices of the rectangle.
Write down the co-ordinates of A, B, C and D.

$A($ $\quad$, $\quad$ $),$

$B($ $\quad$, $\quad$ $),$

$C($ $\quad$, $\quad$ $),$

$D($ $\quad$, $\quad$ $)$

(c) Find the co-ordinates of the midpoint of $[AC]$.

(d) Use the distance formula to find $|AC|$.

(e) Find the slope of DB.

(f) Use your answer to part **(e)** to find the equation of the line DB.

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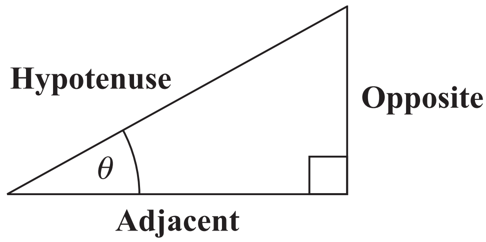
Question 13 **(Suggested maximum time: 10 minutes)**

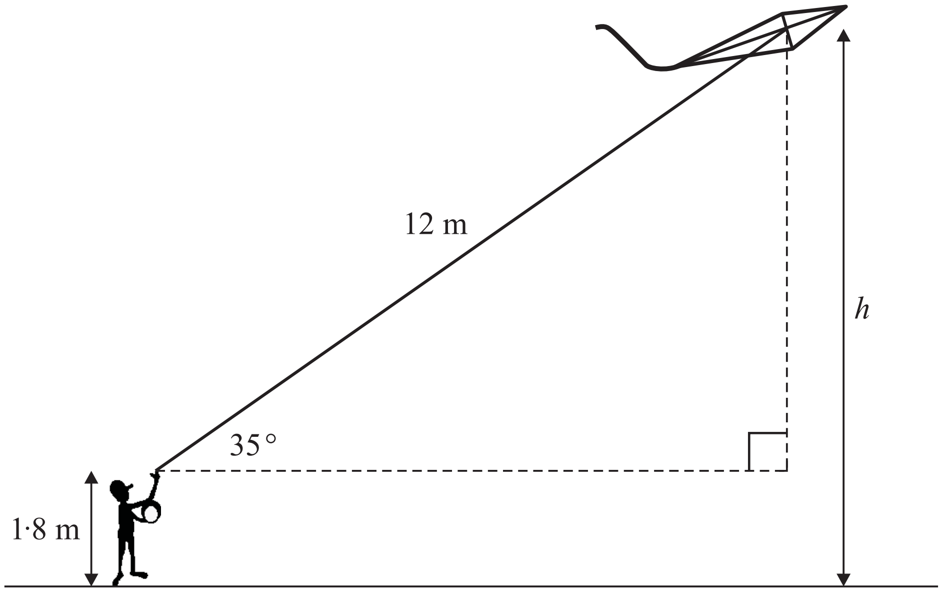
(a) The diagram below shows the angle θ in a right-angled triangle.



Fill in the appropriate words given in the diagram to complete the trigonometric ratios below.

$\sin\theta = \dfrac{\qquad}{\text{hypotenuse}}$ $\quad\cos\theta = \dfrac{\qquad}{\qquad}$ $\quad\tan\theta = \dfrac{\text{opposite}}{\qquad}$

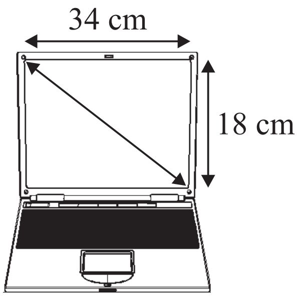
(b) Paul wants to find out how high his kite is above ground level. He measures the angle of elevation of the top of the kite using a clinometer. The angle is 35°. The length of string released from his hand to the kite is 12 m. The string is held 1·8 m above ground level, as shown in the diagram.



Use Paul's measurements to find h, the height of the kite above ground level. Give your answer correct to one decimal place.

(c) The size of a laptop screen is defined by the diagonal measurement of the screen, rounded to the nearest whole number.

A laptop screen has a width of 34 cm and a height of 18 cm, as shown.



(i) Use the Theorem of Pythagoras to calculate the size of the laptop screen.

(ii) The sizes of screens are commonly advertised in inches, where 1 inch = 2·54 cm. Use your answer to part **(i)** to find the size of the laptop screen in inches.

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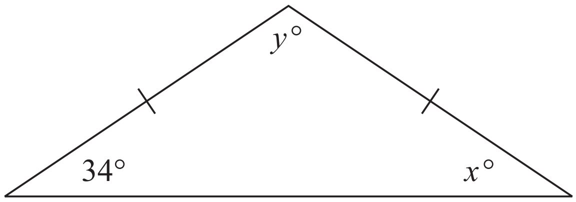
Question 14 **(Suggested maximum time: 5 minutes)**

(a) The diagram shows an isosceles triangle.



(i) Find the measure of the angle x°.

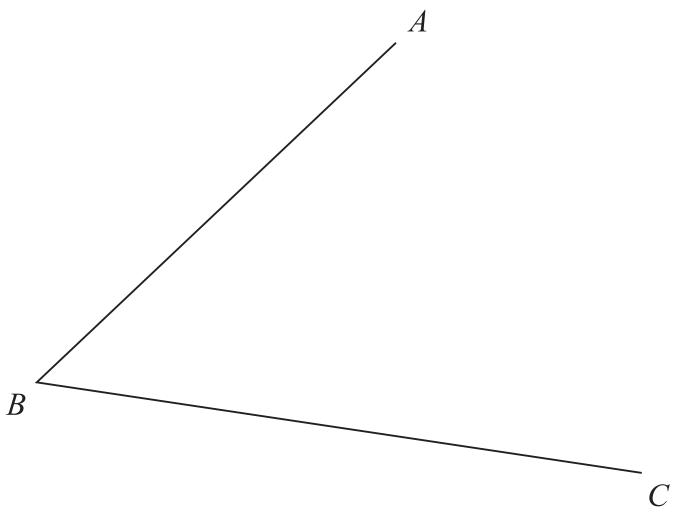
$x^\circ =$

(ii) Find the measure of the angle y°.

$y^\circ =$

(iii) Write down one property of an isosceles triangle.

(b) Using only a compass and a straight edge, construct the bisector of the angle ABC.
Show all construction lines clearly.



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Pre-Junior Certificate, 2015 – Ordinary Level

Mathematics – Paper 2

Time: 2 hours

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